EnviroLearn: A Gamified Environmental Education Platform for Schools

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***Abstract*—Environmental degradation driven by climate change, pollution, and resource depletion demands urgent educational interventions at the school level. Traditional environmental education methods suffer from low interactivity and inadequate student engagement, resulting in limited awareness and participation in sustainable practices. This paper presents EnviroLearn, a gamified web-based platform designed to deliver environmental education to school students through interactive videos, quizzes, educational games, and real-world eco-challenges. The system incorporates gamification elements such as points, badges, leaderboards, and missions to motivate students and sustain engagement. A role-based access control mechanism differentiates the experience for students and teachers, while geo-tagging and image metadata verification ensure the authenticity of submitted environmental activities. Built using HTML, CSS, JavaScript, and Python Flask, EnviroLearn provides an accessible and effective digital tool aligned with SDG 4 (Quality Education) and SDG 13 (Climate Action).**

***Index Terms*—Gamification, Environmental Education, E-Learning, Sustainability, Web Application, Python Flask, Role-Based Access Control, Geo-Tagging**

I. Introduction

Environmental issues such as climate change, pollution, and resource depletion represent some of the most pressing global challenges of the 21st century. The formation of environmentally responsible citizens begins in schools, where attitudes toward sustainability and conservation are shaped at an early age. However, conventional environmental education in schools is often lecture-driven and low in interactivity, leading to reduced student engagement and limited long-term retention of environmental concepts.

Digital learning platforms have proven effective at improving student engagement across many educational domains, and gamification—the application of game-design elements in non-game contexts—has emerged as a particularly powerful strategy to make learning interactive, motivating, and memorable. By rewarding students for completing learning activities and environmental tasks, gamified platforms can convert passive recipients into active participants who are intrinsically motivated to pursue eco-friendly behaviors.

EnviroLearn is a gamified environmental education platform developed to address the shortcomings of traditional environmental education in schools. The platform provides structured learning content through short educational videos on topics such as pollution, climate change, biodiversity, and waste management. Students complete quizzes based on video content and participate in real-world environmental challenges such as waste-sorting or tree-planting activities. They earn points, badges, and levels, and compete on leaderboards—mechanisms that sustain motivation and foster a spirit of environmental responsibility and passion among learners.

From a technical perspective, EnviroLearn is built using HTML, CSS, and JavaScript for the frontend, and Python Flask for the backend. The system incorporates session-based authentication with password hashing, role-based access control (RBAC) for students and teachers, geo-tagging via the browser geolocation API to verify the physical location of activity submissions, and EXIF metadata extraction from uploaded images to validate the authenticity of challenge submissions. Teachers are provided with a dedicated monitoring dashboard that aggregates student activity data and enables oversight of participation and environmental engagement.

The platform aligns with two United Nations Sustainable Development Goals: SDG 4 (Quality Education) and SDG 13 (Climate Action), making it not only a pedagogical tool but also a contribution toward global sustainability objectives.

II. METHODOLOGY

EnviroLearn is designed around five core technical components that together deliver a secure, engaging, and verifiable environmental education experience. The overall system is organized into three tiers—Frontend, Backend, and Database—that communicate in a layered architecture as illustrated in Fig. 1. The frontend handles all user-facing interactions including login, video learning, quizzes, educational games, and environmental challenge submissions. The backend processes and validates all incoming data: it evaluates quiz responses, computes game scores, manages the reward system, and verifies geo-tagged and image-based challenge evidence. The database tier persistently stores all user account information, scores, reward points, challenge records, and ac-

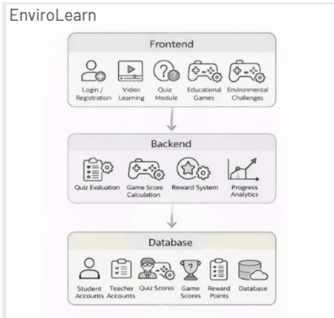

Fig. 1. System architecture of the EnviroLearn platform.

tivity logs. This separation of concerns ensures maintainability, scalability, and security across all platform operations.

A. User Authentication

Technical Method: Session-Based Authentication with Password Hashing
EnviroLearn uses secure authentication mechanisms to verify users during login. Passwords are stored using cryptographic hashing techniques, and upon successful verification, authenticated sessions are created to grant secure access to platform features. This ensures that only registered users can interact with learning content, quizzes, games, and challenge submissions.

B. Role-Based Access Control (RBAC)

Technical Method: Role-Based Access Control (RBAC)
Users are categorized into two roles—Students and Teachers—with access permissions controlled accordingly. Students can browse video learning modules, attempt quizzes, play educational games, participate in environmental challenges, and view their accumulated points, badges, and leaderboard rankings. Teachers, on the other hand, are granted access to an administrative dashboard that displays student submissions, activity records, and verification details, enabling them to monitor participation and assess environmental engagement across their class.

C. Geo-Tagging Verification

Technical Method: Browser Geolocation API Integration
To ensure that environmental challenge activities are genuinely performed in the field rather than fabricated remotely, EnviroLearn retrieves the user's live geographic coordinates using the browser's built-in geolocation functionality. When a student submits an environmental activity, their current coordinates are recorded and stored alongside the submission. This geo-tagging mechanism helps verify that activities such as waste collection, tree planting, or water conservation tasks are carried out at a legitimate physical location.

D. Image Metadata Processing

Technical Method: EXIF Metadata Extraction
Students are required to upload photographic evidence when submitting environmental challenge completions. Uploaded images are analyzed to extract embedded EXIF metadata, including GPS coordinates and timestamps. This information is cross-referenced with the geo-tagged submission data to validate the authenticity of challenge submissions and prevent fraudulent reporting.

E. Teacher Monitoring Dashboard

Technical Method: Data Aggregation
Teachers access a centralized dashboard that aggregates and displays student submissions, quiz scores, game scores, reward points, activity records, and geo-verification details. The dashboard provides teachers with a holistic view of class-wide environmental engagement, enabling them to identify highly active students, recognize achievement patterns, and intervene where participation is low. This data-driven approach supports informed instructional decisions and reinforces a culture of environmental accountability within the classroom.

F. System Implementation Phases

The EnviroLearn system is structured into three implementation phases:

Frontend Phase:

- Provides Login and Registration for both students and teachers.
- Includes video learning modules presenting short educational content on environmental topics.
- Offers a quiz module where students answer questions based on the video content they have watched.
- Features educational games that reinforce environmental concepts through interactive and fun activities.
- Presents environmental challenges that motivate students to take real-world sustainability actions.

Backend Phase:

- Automatically evaluates quiz answers and calculates scores.
- Processes game interactions and computes game scores.
- Manages the reward system, awarding points, badges, and level progressions.
- Generates performance analytics and progress reports accessible to teachers.
- Validates geo-tagged coordinates and EXIF metadata from challenge submissions.

Database Phase:

- Stores student and teacher account information securely.
- Saves quiz scores, game scores, and reward points earned by each user.
- Records challenge submission data including images, geo-coordinates, and timestamps.
- Maintains the complete operational database for all system modules.

III. LITERATURE REVIEW

Dicheva et al. [1] conducted a systematic review of gamification in educational contexts and found that incorporating game elements such as points, badges, and leaderboards significantly increases student motivation and engagement. Their work established a foundational framework for understanding how gamification mechanisms can be effectively applied to formal education settings.

Hamari et al. [2] performed an empirical meta-analysis of gamification studies and demonstrated that gamified systems produce positive effects on learning outcomes, particularly when the gamification elements are well-aligned with the educational objectives. The study also highlighted the importance of social comparison features such as leaderboards in sustaining competitive motivation.

Scobell and Lee [3] examined the integration of digital game-based learning in environmental science classrooms and reported marked improvements in students' knowledge retention and pro-environmental attitudes compared to traditional instruction. Their findings support the use of game-based approaches specifically for sustainability education.

Kuo and Chuang [4] investigated the effects of a gamified e-learning platform on elementary school students and found that students who engaged with gamified content demonstrated higher levels of intrinsic motivation and completed learning modules at significantly higher rates than those using conventional online resources. The study underscored the role of immediate feedback and reward systems in reinforcing desired behaviors.

Yildiz Durak [5] explored the use of flipped classroom models combined with gamified quizzes in secondary schools and reported that students exhibited improved conceptual understanding and greater willingness to participate in classroom discussions. This aligns with EnviroLearn's video-first, quiz-follow approach to content delivery.

Sung and Hwang [6] proposed a collaborative game-based learning environment for science education and found that students who participated in collaborative challenges developed stronger critical thinking skills and deeper content understanding than those engaged in individual study. EnviroLearn's leaderboard and mission features similarly encourage collaborative environmental action.

Manzano-León et al. [7] reviewed studies on gamification applied to sustainability education and concluded that platforms combining informational content with gamified rewards are effective at shifting student attitudes toward environmentally responsible behavior. This directly supports EnviroLearn's dual approach of education and behavioral motivation.

Fauzi et al. [8] evaluated a mobile-based environmental education application for middle school students and reported that interactive digital content significantly outperformed print-based materials in both engagement and post-test environmental knowledge scores. Their work highlights the advantage of digital platforms in resource-constrained educational environments.

Huang and Soman [9] provided a practitioner's guide to gamification in education, identifying five critical steps for successful implementation: understanding the target audience, defining learning objectives, structuring experience points, applying leaderboards, and ensuring progression with levels and badges. EnviroLearn's design directly reflects these principles, incorporating all five elements tailored to school-level environmental education.

Caponetto et al. [10] analyzed the relationship between serious games and environmental awareness campaigns, finding that game-based interventions produced lasting changes in participants' daily behaviors, including recycling rates and energy conservation habits. This evidence supports the use of EnviroLearn's real-world environmental challenges as a mechanism for sustained behavioral change beyond the classroom.

Stott and Neustaedter [11] examined the role of transparency and fairness in gamified systems, arguing that clearly communicated reward structures and verifiable achievement criteria are essential for maintaining student trust and long-term participation. EnviroLearn's geo-tagging and EXIF metadata verification mechanisms address this concern by ensuring that challenge rewards are grounded in verified real-world activity.

Hanus and Fox [12] conducted a semester-long study comparing gamified and non-gamified courses and found that while short-term engagement increased with gamification, long-term outcomes depended critically on the intrinsic relevance of the gamified tasks to students' values. EnviroLearn addresses this by grounding all gamified tasks in genuine environmental issues that are locally and globally relevant to students' lived experience.

IV. CONCLUSION

EnviroLearn demonstrates that gamification can be an effective strategy for transforming environmental education in schools from a passive, lecture-based experience into an engaging, interactive, and behaviorally impactful learning journey. By combining short educational videos, knowledge-testing quizzes, educational games, and real-world geo-verified environmental challenges with a robust reward system of points, badges, levels, and leaderboards, the platform sustains student motivation while deepening environmental awareness across topics such as pollution, climate change, biodiversity, and waste management.

The technical architecture—built on HTML, CSS, JavaScript, and Python Flask—ensures accessibility and ease of deployment. The incorporation of session-based authentication, role-based access control, browser geolocation verification, and EXIF image metadata processing ensures the security, integrity, and authenticity of all platform interactions and student submissions. The teacher monitoring dashboard provides educators with the data-driven insights needed to support and guide student participation effectively.

EnviroLearn contributes directly to SDG 4 (Quality Education) by delivering structured, technology-enhanced environmental education, and to SDG 13 (Climate Action) by raising student awareness of climate-related issues and motivating pro-environmental behaviors. Future work will focus on expanding the platform's content library, integrating AI-driven adaptive learning pathways, and conducting longitudinal studies to assess the long-term impact of gamified environmental education on students' sustainability practices.

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